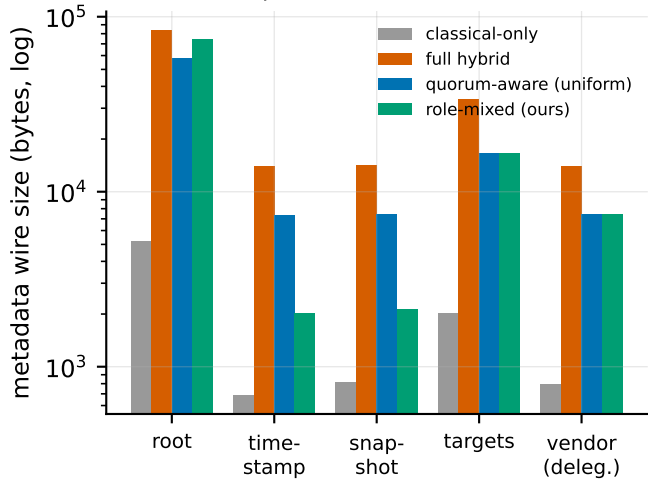


(a) per-role metadata size



(b) ECU per-update critical path (root amortized)

